

Roshan Siddhartha Sivakumar

615-364-5138 | roshan.siddhartha.24@gmail.com | [linkedin.com/in/roshansiddhartha](https://www.linkedin.com/in/roshansiddhartha) | github.com/rroshann | rroshann.github.io

PROFESSIONAL SUMMARY

Data scientist with hands-on experience in machine learning, LLM applications, and production data pipelines. Fine-tuned a 14B reasoning LLM for an AllianceBernstein quantitative research capstone, built an AI chatbot presented to Nissan North America executives, and engineered BI automation eliminating 40+ hours of weekly manual work.

EDUCATION

Vanderbilt University — Data Science Institute

M.S. in Data Science, GPA: 3.88/4.0

Nashville, TN

Aug. 2024 – May 2026

Vellore Institute of Technology

B.Tech in Mechanical Engineering, GPA: 8.47/10.0

Vellore, India

Jul. 2019 – Jun. 2023

EXPERIENCE

Data Science Intern

Aug. 2025 – Present

SERVPRO

Lebanon, TN

- Built a multimodal AI agent using GPT-4 Vision to autonomously navigate legacy enterprise portals, achieving 99% scraping reliability with spatial reasoning-based popup detection (Playwright)
- Architected two BI automation pipelines processing 13,500+ job records nightly into executive reports with 20+ KPIs and 11 pivot tables, eliminating 40+ hours of weekly manual work
- Productionalized all pipelines via GitHub Actions with Azure AD OAuth 2.0, Microsoft Graph API delivery, and SharePoint upload for 100% zero-touch operation

AI Intern, Leadership Development Program

Jun. 2025 – Aug. 2025

Vanderbilt University, Owen Graduate School of Management

Nashville, TN

- Built a secure web app to automate coach-student matching, cutting manual work from days to under 10 minutes; designed an anonymized data pipeline generating matches and rationale via OpenAI API
- Developed an embedded AI assistant with safety guardrails, refusal logic, and prompt evaluation to support business students

PROJECTS

Reasoning-Augmented Factor Extraction from SEC Filings | *PyTorch, QLoRA, GRPO*

Jan. 2026 – Present

- Architected an end-to-end NLP pipeline extracting 67,741 thematic factors from 2,542 SEC filings (10-K/10-Q) across 86 tickers for an AllianceBernstein-sponsored capstone, producing tradeable sentiment signals validated against 21-day excess returns
- Fine-tuned a 14B-parameter reasoning LLM (DeepSeek-R1) using QLoRA + GRPO reinforcement learning with a custom composite ordinal reward, lifting macro F1 by +12.7% and producing a long-short portfolio with IR 2.93
- Engineered a 48-cell quantitative backtest grid with out-of-sample validation on 605 held-out post-2023 filings (OOS IR 1.99); delivered a 51-page technical report and final presentation to AllianceBernstein

FabOps Copilot: Agentic Stockout Risk Diagnosis | *LangGraph, RAG, AWS, MLflow*

Jan. 2026 – Apr. 2026

- Architected a 9-node LangGraph agent diagnosing semiconductor service-part stockout risk across 200 parts and 9 fabs, achieving 83.3% task success via cross-family LLM-as-judge evaluation
- Engineered a RAG pipeline over 1,079 SEC filing chunks with inline citations; deployed on a serverless AWS stack (Lambda, DynamoDB, API Gateway) with CI gating an 18-case gold eval on every push
- Implemented Croston intermittent-demand forecasting on a 2,674-part benchmark, logging sMAPE to MLflow

Nissan Dealer Training Analytics & AI Chatbot | *RAG, Amazon QuickSight, Web Scraping*

Aug. 2025 – Dec. 2025

- Analyzed 140,000+ logs to diagnose low engagement; led development of a RAG-based chatbot prototype for the 2026 Nissan Leaf integrating web-scraped specs, official documentation, and competitor data
- Presented findings and prototype to Nissan NA executives, informing a solution for North American dealer rollout

AI-Generated Image Detection | *PyTorch, VGG16, EfficientNet, Grad-CAM*

Oct. 2025 – Dec. 2025

- Led a deep learning project detecting AI-generated images, achieving 97.94% accuracy and 0.998 ROC-AUC on 120K images with VGG16 transfer learning; benchmarking found EfficientNet-B0 reached 98.42% accuracy with 26× fewer parameters
- Implemented Grad-CAM interpretability, revealing synthetic texture artifacts vs. natural object boundaries

TECHNICAL SKILLS

Languages: Python (Pandas, NumPy, Scikit-learn, PyTorch), SQL (MySQL)

ML & AI: Deep Learning (CNNs, Transfer Learning), NLP, RAG, LLM Fine-tuning (QLoRA, GRPO), LLM APIs (OpenAI, Claude, Gemini), Multi-Agent Systems (LangGraph, LangChain), Computer Vision (YOLOv8, Grad-CAM, OpenCV), MLOps (MLflow)

Platforms & Tools: AWS (Lambda, S3, API Gateway), Docker, GitHub Actions, MongoDB, DynamoDB, PySpark, Tableau, PowerBI, Streamlit, Flask, Playwright, Azure AD, Git